Partial Differential Equations (PDE)
Assignment 4

[Fourier Series]

(a) Let $\phi(x) = x$ for $0 \leq x \leq l$. Calculate its Fourier sine series and cosine series.

(b) Find the Fourier cosine series of the function x^2, and set $x = 0$ to find the sum of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$.